

Introduction to Derivatives

Module: Derivatives Basics

Chapter: 01 - Introduction to Derivatives

Level: Beginner

Learning Objectives

After completing this chapter, you will be able to:

- Define a derivative.
 - Understand why derivatives exist.
 - Differentiate between the underlying asset and the derivative.
 - Identify the major types of derivatives.
 - Explain how derivatives are used in financial markets.
 - Understand the benefits and risks of derivatives.
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What is a Derivative?

A **derivative** is a financial contract whose value is **derived from (depends on)** the value of another asset.

The original asset is called the **Underlying Asset**.

A derivative itself has **no independent value**.

Its price changes whenever the price of the underlying asset changes.

Simple Definition

A derivative is a contract between two or more parties whose value depends on an underlying asset.

What is an Underlying Asset?

An **underlying asset** is the asset from which a derivative gets its value.

Examples include:

Asset Type	Examples
Stocks	Reliance, TCS, Infosys

Asset Type	Examples
Indices	Nifty 50, Sensex
Commodities	Gold, Silver, Crude Oil
Currency	USD/INR, EUR/USD
Interest Rates	Government Bonds
Agricultural Products	Wheat, Cotton

Basic Idea

Imagine wheat costs **₹2,000 per quintal** today.

A bakery worries that wheat prices may rise next month.

The bakery signs an agreement today to buy wheat after one month at **₹2,050**.

This agreement is **not wheat itself**.

It is a **contract based on wheat**.

That contract is a **derivative**.

How Derivatives Work



Whenever the price of the underlying asset changes, the derivative's value also changes.

Real-Life Example

Suppose:

- Current Gold Price = ₹60,000 per 10g
- You believe prices will rise.

Instead of buying physical gold, you purchase a ****Gold Futures Contract****.

Scenario 1

Gold rises to ₹63,000.

Your futures contract also increases in value.

Profit ✓

Scenario 2

Gold falls to ₹58,000.

Your contract loses value.

Loss ✗

Why Were Derivatives Created?

Originally, derivatives were created to reduce business risk.

Farmers wanted protection against falling prices.

Manufacturers wanted protection against rising prices.

Importers wanted protection against currency fluctuations.

Exporters wanted stable exchange rates.

Over time, derivatives also became investment and trading instruments.

Major Uses of Derivatives

1. Hedging

Protect against future price movements.

Example:

- Airline hedging fuel prices
- Exporter hedging USD/INR rate

2. Speculation

Attempt to earn profit by predicting price movement.

Example:

Trader believes Nifty will rise and buys Nifty Futures.

3. Arbitrage

Profit from price differences in different markets.

Example:

Gold trades cheaper in one exchange than another.

Professional traders exploit the difference.

4. Portfolio Management

Large investors use derivatives to reduce portfolio risk.

Example:

Mutual funds hedge market risk using index futures.

Types of Derivatives

There are four major derivative contracts.

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Derivatives

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├── Forward

├── Futures

├── Options

└── Swaps

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## 1. Forward Contract

- Private agreement
- Customized
- Not traded on exchanges
- Higher counterparty risk

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## 2. Futures Contract

- Standardized agreement
- Traded on exchanges
- Highly regulated

- Daily settlement

Examples:

- Nifty Futures
- Gold Futures
- Crude Oil Futures

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### 3. Options Contract

Provides the **right**, but **not the obligation**, to buy or sell an asset.

Two types:

- Call Option
- Put Option

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### 4. Swaps

Agreement to exchange cash flows.

Common examples:

- Interest Rate Swap
- Currency Swap

Mostly used by banks and large corporations.

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## Exchange-Traded vs OTC Derivatives

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| Feature           | Exchange Traded | OTC   |
|-------------------|-----------------|-------|
| Regulation        | High            | Low   |
| Standardized      | Yes             | No    |
| Counterparty Risk | Low             | High  |
| Transparency      | High            | Low   |
| Liquidity         | High            | Lower |

Examples:

Exchange:

- NSE
- BSE
- CME

OTC:

- Banks
- Financial Institutions

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## Participants in Derivatives Market

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| Participant   | Purpose                 |
|---------------|-------------------------|
| Hedgers       | Reduce Risk             |
| Speculators   | Earn Profit             |
| Arbitrageurs  | Price Difference Profit |
| Market Makers | Provide Liquidity       |

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## Advantages of Derivatives

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- Risk management
  - Price discovery
  - High liquidity
  - Portfolio diversification
  - Leverage
  - Lower transaction costs
  - Efficient capital usage
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## Risks of Derivatives

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- High leverage can magnify losses.
  - Market volatility.
  - Counterparty risk (OTC).
  - Liquidity risk.
  - Complexity.
  - Regulatory risk.
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## Advantages vs Risks

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| Advantages | Risks         |
|------------|---------------|
| Hedging    | Leverage Risk |

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| Advantages         | Risks                            |
|--------------------|----------------------------------|
| Liquidity          | Volatility                       |
| Capital Efficiency | Complexity                       |
| Price Discovery    | Counterparty Risk                |
| Diversification    | Unlimited Loss (some strategies) |

## Common Examples

| Derivative        | Underlying     |
|-------------------|----------------|
| Nifty Futures     | Nifty 50       |
| Gold Futures      | Gold           |
| USDINR Futures    | USD/INR        |
| Reliance Option   | Reliance Stock |
| Crude Oil Futures | Crude Oil      |

## Key Terminology

| Term             | Meaning                                       |
|------------------|-----------------------------------------------|
| Derivative       | Contract whose value depends on another asset |
| Underlying Asset | Asset on which derivative is based            |
| Futures          | Standardized exchange-traded contract         |
| Forward          | Private customized contract                   |
| Option           | Right, not obligation                         |
| Swap             | Exchange of future cash flows                 |
| Hedging          | Reducing risk                                 |
| Speculation      | Taking risk for profit                        |
| Arbitrage        | Profit from price differences                 |

## Summary

- A derivative derives its value from another asset.
- The underlying asset determines the derivative's price.

- The four major derivative contracts are:
    - Forward
    - Futures
    - Options
    - Swaps
  - Derivatives are mainly used for:
    - Hedging
    - Speculation
    - Arbitrage
  - They are powerful financial tools but involve significant risk if not used properly.
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## Quick Revision

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- ✓ Derivative = Contract
  - ✓ Underlying = Original Asset
  - ✓ Futures = Obligation
  - ✓ Options = Right, not obligation
  - ✓ Swaps = Exchange of cash flows
  - ✓ Hedging = Reduce risk
  - ✓ Speculation = Profit from price movement
  - ✓ Arbitrage = Profit from price differences
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## What's Next?

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In the next chapter, you'll learn:

- How derivatives originated.
- Evolution from agricultural markets to modern financial exchanges.
- Growth of derivatives in India and globally.
- Role of NSE, BSE, CME, and other major exchanges.